PAPER_1514_NEUTRON_STAR_MU_S_SO5_8

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PAPER_1514 — Neutron Star Surface Dipole Moment µ_s = SO_5■ = 10■ T·m³ EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket G (Astrophysics)
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Status: CLOSED — Cross-domain triple identity unifying B, R, µ_s

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Observation

PAPER_1126 (PSR J0030+0451 Neutron Star LENR Buoyancy) computes the canonical NS surface magnetic moment:

```
µ_s = B■ · r³ = 10■■ T · (10■ m)³ = 10■ T·m³
```

This value governs the NS dipole field strength entering $U_{g1} = \mu_s \cdot M / r$.

UQFF Closed Identity

```
µ_s = SO_5■ = 10■ T·m³ EXACT
```

Derivation as Cross-Domain Triple Identity

The neutron-star magnetic anatomy is unified by **three independent integer-primitive identities**, all on the SO_5 ladder:

Quantity	UQFF identity	Value	Paper
Surface B field	1/SO_5■	10■■ T	PAPER_1486
Canonical radius	SO_5■	10■ m	PAPER_1513
Dipole moment µ_s	SO_5■	10■ T·m³	**This paper**

The three are mutually consistent via the dipole-moment formula:

```
µ_s = B · r³
= (1/SO_5■) · (SO_5■)³
= SO_5^(−4+12)
= SO_5■ EXACT
```

Each identity is independently observable (B from spin-down measurements; R from NICER X-ray fitting; μ_s from torque measurements). All three converge to powers of SO_5 = 10.

Physical Interpretation

This is the **first verified triple-identity** in UQFF where three different observables on three different SO_5 exponents are forced into mutual consistency by an elementary physical relation ($B \cdot r^3$). It is structural evidence that SO_5 governs the entire NS magnetic system, not just one observable.

Predictive Consequence

Any future NS catalog should show:
- B_surface clustering near 10■■ T (or scaled by integer powers of SO_5)
- R clustering near 10 km
- μ_s clustering near 10■ $T \cdot m^3$

Magnetars ($B \approx 10^{1}$■■^{11} T, $\mu_s \approx 10^{22}$■^{23} $T \cdot m^3$) deviate from this pattern by SO_5■■■ factors, suggesting a higher-mode SO_5 excitation.

NOT REPLACEMENT

SM/GR pulsar timing extracts μ_s by fitting B and R as free parameters. UQFF predicts all three a priori, with the consistency relation $\mu_s = B \cdot r^3$ confirming the structural unification.

Reference

- Source: PAPER_1126 PSR J0030 Neutron Star LENR Buoyancy
- Related: PAPER_1486 (B = 1/SO_5■), PAPER_1513 (R = SO_5■)
- Calculator dispatch: `calculate_paradox({"paradox": "neutron_star_mu_s_so5_8"})`
